2nd Electronic Circuits and Systems Conference
September 6-8, 1999, Bratislava, Slovakia

A 32-bit RISC Microprocessor with DSP Functionality

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Abstract. We have designed a 32-b RISC microprocessor with 16-/32-b fixed-point DSP functionality. This processor, called YD-RISC, has functional units for arithmetic operation, DSP and memory accesses. They operate in parallel in order to remove stall cycles after DSP or load/store instructions with one or more latency cycles. High performance is achieved with the parallel functional units while adopting a sophisticated five-stage pipeline structure. The DSP unit can execute one 32-b multiply-accumulate or 16-b complex multiply instruction every one or two cycles through two 17-b $\times$ 17-b multipliers and operand examination logic circuits. Power-saving circuits allow low power consumption.

I. Introduction

Embedded microcontrollers are used to execute general-purpose programs [1], [2]. DSPs are used for specialized applications such as image and voice processing [3]. Recently low-cost embedded processors with both microcontroller and DSP functionality have become necessary in the advanced consumer electronics applications. Simply combining both functions using two separate cores in a single chip is not cost-effective and not appropriate for embedded systems considering the low-cost need in consumer electronics [4]. Therefore, we designed a processor that combines both microcontroller and DSP functions into one processor without doubling resources, while providing the main features of RISC architectures for flexibility in programming, also achieving high DSP performance.

Design methodology, which is mainly classified into full-custom design and logic synthesis using gate-level cell libraries, is a very important point for successful processor designs. The full-custom method is suitable for high performance, small area and low power consumption, but it is time-consuming and labor-intensive. The method through synthesis is time and labor-saving, thus making short time-to-market possible which is an important requirement for successful marketing. Moreover, synthesis makes it easy to improve designs, because updating designs is simply re-synthesizing using improved libraries and process technology. We designed the YD-RISC through logic synthesis and automatic place-and-route and achieved a high clock frequency by refining the architecture with the five-stage pipeline.

II. Design Points

In the design of the YD-RISC, various new concepts are used to optimize performance, area and power consumption. These concepts are described in detail below.

A. Pipeline and Parallelism

Instructions are divided into ALU, DSP and load/store instructions. As shown in Fig. 1, the ALU pipeline consists of five stages in accordance with the basic RISC pipeline structure, and DSP and load/store pipelines have separate and parallel execution stages independent of ALU execution and write-back stages. This enables the instructions to be executed continuously after the DSP or load/store instructions with one or more latency cycles regardless of the completion of these instructions as shown in Fig. 2, thus improving performance dramatically.

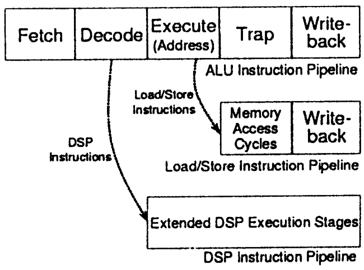

Fig. 1. Basic pipeline structure.

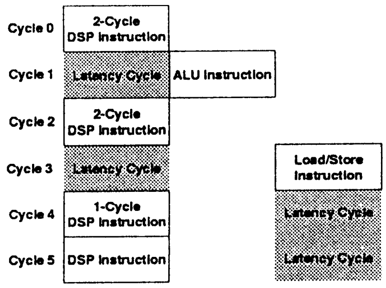

Fig. 2. Parallelism between functional units.

B. DSP Unit

The DSP unit can do both 16-b × 16-b and 32-b × 32-b multiplications. A 32-b × 32-b multiplier occupies too large area and can perform only one 16-bit multiplication at one time. In contrast, two 16-b × 16-b multipliers occupy about half the area of one 32-b × 32-b multiplier and can perform two 16-b × 16-b multiplications simultaneously. The DSP unit of the YD-RISC can execute two independent 16-b × 16-b multiplications in one cycle and one 32-b × 32-b multiplication in two cycles using the two 17-b × 17-b multipliers. In addition, the number of cycles in 32-b × 32-b multiplication is further reduced to only one through the operand examination scheme as shown in Tab. I.

Tab. I. Operand examination scheme for 1-cycle 32-bit multiplication

Operation	Operand condition for 1-cycle 32-bit multiplication
32-bit signed multiply	Operand[31:16] is equal to $0000_{(16)}$ or $FFFF_{(16)}$ or Operand[15:0] is equal to $0000_{(16)}$.
32-bit unsigned multiply	Operand[31:16] is equal to $0000_{(16)}$ or Operand[15:0] is equal to $0000_{(16)}$.

Tab. II. Average instruction length

Program	No. of executed instructions	Average instr. length
hello	1096	24.44
hanoi4	3021	25.41
bsort10	28160	26.64
ssort10	20455	23.27
qsort10	15425	24.35
siv1024	22044	21.84
dct22f	173583	21.20
fft8f	15389	21.41
dct22d	2381	22.28
fft8d	508965	22.98
fir	40653	24.74
iir	49362	22.37

C. Instruction Prefetch

This architecture has no instruction or data cache, and thus can be vulnerable to execution stall due to long memory access time and insufficient instruction availability. In order to overcome this drawback, the processor adopts an internal 4-Kbyte SRAM and a prefetch buffer, which is implemented as a circular buffer with a head pointer and a tail pointer. The loop control of the prefetch buffer reduces the bus utilization of this processor, and thus improves performance when the program consists of small loops to be processed repeatedly. Moreover, this architecture adopts a variable-length instruction format of 16 bits, 32 bits, or 48 bits, which reduces the average instruction length below 25 bits in most programs as shown in Tab. II. Improved code density due to this variable-length instruction format not only saves memories, but also improves instruction supply and performance. Also, bus arbitration between instruction memory access and data memory access is optimized using the results through various simulations.

D. Power Consumption

Low power nature is essential in embedded processors. In power-down mode provided in our processor, only the clocks for the logic related to the refresh, interrupt, timer and reset are enabled, while all other clocks are disabled, using the circuit shown in Fig. 3. Also, power consumption by unnecessary switching in execution blocks is avoided through the circuit in Fig. 4, which permits only enabled execution blocks to receive new input operands. Power consumption is noticeably reduced by these power-saving techniques.

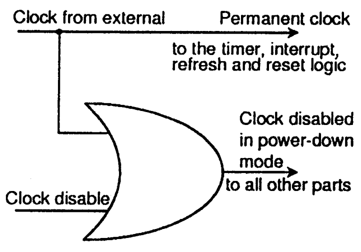

Fig. 3. Clock disable circuit.

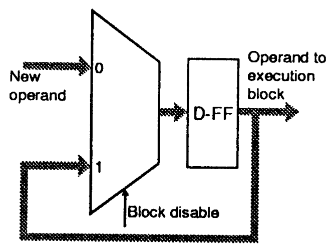

Fig. 4. Execution block disable.

III. Implementation and Performance

The block diagram in Fig. 5 shows the main architecture of the entire microprocessor. The logical model of this processor is described by using an industry-standard HDL and is verified with a HDL software simulator. The register file, prefetch buffer and internal SRAM are designed with established macro cell libraries, while the others are logic-synthesized with gate-level cell libraries. The layout of the processor is shown in Fig. 6 and measures about 8mm × 8mm. The processor designed in 3.3V 0.6-um triple metal technology shows the clock frequency of 50 MHz. This frequency is high considering that the frequency of LG Semicon's custom-designed GMS30C32132 adopting the similar architecture and the same technology is 30MHz [5]. Moreover, the frequency of this processor can be almost doubled by redesign using current available 0.35-um technology. The YD-RISC shows slightly better performance than GMS30C32132 in integer applications and twice as high performance as GMS30C32132 in DSP applications. Table III shows the complex FFT performance comparison of the YD-RISC with different DSP processors, assuming that the performance of the YD-RISC is one.

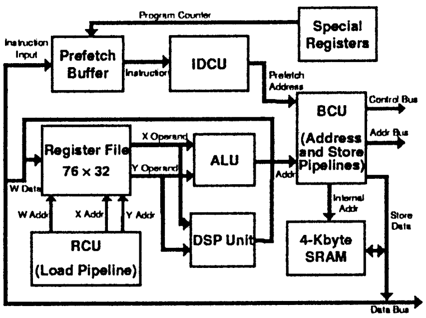

Fig. 5. Architecture of the YD-RISC.

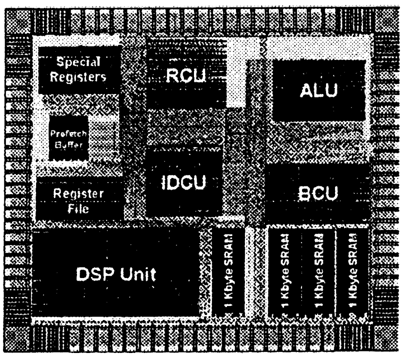

Fig. 6. Layout of the YD-RISC.

Tab. III. 1K 16-bit complex FFT performance comparison of the YD-RISC with different processors

DSP processor	Relative performance
YD-RISC	1.00
LGS GMS30C32132	0.54
TI TMS320C50	0.31
ADSP 2175	0.68
Motorola 56002	0.78
AT&T DSP1617	0.30
NEC uP77016	0.74

IV. Conclusions

In this microprocessor, RISC and DSP functionality are combined cost-effectively, while high performance is achieved. Excellent performance on 32-b × 32-b multiplication and simultaneous 16-b × 16-b multiplications is achieved through two 17-b × 17-b multipliers and operand examination logic circuits. Power-down mode and disabling execution blocks reduce the power consumption. By designing with logic synthesis and established libraries, we reduced design time as well as achieved a high clock frequency by refining the processor architecture.

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